NMC-Opt Studio: Installation, Configuration & Execution Guide

Complete Technical Operations Manual for Local Deployment and CLI Pipeline Execution

Target Audience: Software Engineers, Academic Evaluators, and Compiler Researchers
Repository URL: https://github.com/MSN-1307/NMC-OPT-CD | **Platform Compatibility:** Windows / Linux / macOS
Prerequisites: Python 3.10+, Node.js 18+, C Compiler (gcc/clang), and Git

1. System Prerequisites

Before initializing NMC-Opt Studio, verify that the following runtime environments are installed:

- **Python:** Version 3.10 or higher (Python 3.11, 3.12, 3.13, and 3.14 verified). Ensure pip is updated.
- **Node.js:** Version 18.0 or higher with npm (Node.js 20+ LTS recommended).
- **C Compiler:** GCC (MinGW for Windows, GNU GCC for Linux) or Clang for native C kernel compilation audits.
- **Git:** Version 2.30+ for repository cloning and version control.

2. Step 1: Repository Cloning & Directory Inspection

Clone the remote Git repository and navigate into the root project directory:

```
git clone https://github.com/MSN-1307/NMC-OPT-CD.git
cd NMC-OPT-CD
```

3. Step 2: Backend Configuration & Server Launch

The backend is built with FastAPI and runs the analytical cost modeling, pycparser AST profiling, and SSA IR generation passes.

```
# Navigate to backend folder
cd backend

# Create and activate virtual environment (Windows PowerShell)
python -m venv venv
.\venv\Scripts\Activate.ps1

# Install Python dependencies and launch server
pip install -r requirements.txt
python -m uvicorn app.main:app --host 127.0.0.1 --port 8008 --reload
```

Verification Endpoints:

- Swagger / OpenAPI Docs: http://localhost:8008/docs
- API Health Endpoint: http://localhost:8008/api/health

4. Step 3: Frontend Installation & Vite Development Server

In a separate terminal, initialize the React / TypeScript user interface:

```
# Navigate to frontend folder
cd frontend

# Install npm dependencies
npm install

# Start Vite development server
npm run dev
```

Open your web browser and navigate to: **http://localhost:5173**

5. Step 4: Standalone Command-Line Interface (CLI) Execution

NMC-Opt Studio includes a standalone CLI utility (backend/cli.py) enabling direct terminal evaluation without opening a browser:

```
# Evaluate Vector Addition with 2,000,000 elements
python backend/cli.py --benchmark vector_add --size 2000000

# Evaluate Matrix-Vector Multiplication (2048 x 2048 matrix)
python backend/cli.py --benchmark gemv --size 4194304 --rows 2048 --cols 2048

# Evaluate arbitrary custom C loop kernel file
python backend/cli.py --custom path/to/my_kernel.c --size 1000000

# Emit raw machine-readable JSON for automated scripts
python backend/cli.py --benchmark relu --size 5000000 --json
```

6. Step 5: Automated Testing & Build Verification

Verify algorithmic correctness across compiler modules using the automated pytest test suite:

```
# Run backend pytest suite (all 8 tests)
python -m pytest backend/tests -v

# Run frontend TypeScript typecheck and production build
cd frontend
npm run build
```

7. Troubleshooting & Common Resolutions

- **Port Conflict on 8008:** If port 8008 is in use, modify the uvicorn start command: python -m uvicorn app.main:app --port 8010 and update frontend API baseUrl.
- **PowerShell ExecutionPolicy:** If Activate.ps1 cannot be loaded, run: Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass.
- **pycparser Syntax Errors:** Ensure uploaded custom C code includes standard function wrappers and avoids compiler-specific assembly extensions.